Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Cylinders and Spheres: Their Shapes and Properties

Look around your home and you will see objects with familiar 3D shapes. A tin of baked beans and a drinking glass are cylinders. A football and an orange are spheres. In this section we look closely at these two shapes and learn how to tell them apart by their properties.

Properties of a cylinder

A cylinder is a solid with two flat circular faces and one curved surface that joins them.

Let us count its parts carefully using a tin can:

• Faces: a cylinder has two faces — the circular top and the circular bottom. The curved part that wraps around the sides is not counted as a flat face; it is a curved surface.

• Edges: an edge is where two faces meet. The curved surface meets each circular face along a circle, so a cylinder has two curved edges (the rim of the top and the rim of the bottom).

• Vertices: a vertex is a corner, where edges meet at a point. A cylinder has no vertices — there is no sharp corner anywhere on it.

• Curved surface: the cylinder has one curved surface joining the two flat circular faces.

So a cylinder has 2 faces, 2 curved edges, 1 curved surface and 0 vertices.

Properties of a sphere

A sphere is a perfectly round solid, like a ball. Every point on its surface is the same distance from its centre.

• Faces: a sphere has no flat faces. Its whole surface is one curved surface.

• Edges: a sphere has no edges. There is no line where two surfaces meet.

• Vertices: a sphere has no vertices. There are no corners at all.

• Curved surface: a sphere has one curved surface that covers the whole shape.

So a sphere has 0 faces, 0 edges, 0 vertices, and 1 curved surface. Because it rolls in every direction, a sphere is sometimes called a "perfectly round" solid.

Comparing the two shapes

Here is a quick way to remember the difference. Both a cylinder and a sphere have one curved surface and no vertices. But a cylinder has two flat circular faces and two curved edges, while a sphere has no flat faces and no edges at all. So when you see a solid with a round flat top and a round flat bottom, it is a cylinder; when you see a solid that is round all over with no flat part, it is a sphere.

A useful check: imagine resting the shape on a table. A cylinder can stand still on its flat circular end, but a sphere will roll away — it has no flat surface to rest on.

Worked example 1

A can of soup has a flat circular top, a flat circular bottom, and a curved side that joins them. How many faces, edges and vertices does it have?

Think: the two flat circular parts are the two faces. The curved side is a curved surface, not a flat face. The curved surface meets the top along a circle and meets the bottom along another circle, giving two curved edges. There are no corners anywhere.

Write: faces = 2, curved edges = 2, vertices = 0.

Worked example 2

Classify each solid as a cylinder or a sphere: (a) a tennis ball, (b) a roll of sticky tape.

Think: a tennis ball is round all over, with no flat face and no edges — it is a sphere. A roll of sticky tape has a flat circular top and a flat circular bottom joined by a curved surface — it is a cylinder.

Write: (a) sphere, (b) cylinder.

Practice

1. How many faces, edges and vertices does a cylinder have?
2. Which of these is a sphere: an ice cream cone, a beach ball, a coin? Explain why.
3. A solid has two flat circular faces and one curved surface. What is the name of this solid?
4. Complete the table for a sphere: faces, edges, vertices, curved surfaces.
5. Name one object in your home shaped like a cylinder and one shaped like a sphere.

Section 2: Cylinders, Spheres, and Finding Volume

Let's meet two more three-dimensional shapes that you see all around you: a cylinder and a sphere.

A cylinder is the shape of a tin of baked beans, a drum, or a drinking glass. Its two ends are identical circles, and the curved surface joining them is smooth and straight along its length. A cylinder has two flat faces (the circular ends), one curved face, and no corners. The line that runs down the middle of the curved face, from the centre of one circle to the centre of the other, is the height (or length) of the cylinder, and the radius of either end is the radius of the cylinder.

A sphere is perfectly round, like a football, a globe, or a marble. It has just one curved face, with no flat faces, no edges and no corners. Every point on the surface of a sphere is exactly the same distance from its centre, and that distance is the radius of the sphere.

So remember: a cylinder has two flat circular faces and one curved face; a sphere has only one curved face and nothing flat at all.

Now we turn to finding how much space a three-dimensional shape takes up. That space is called its volume, and it is measured in cubic units, such as cubic centimetres (cm³).

A cube is a shape whose every face is a square, so all its edges are the same length. If each edge is s units long, then its volume is s × s × s, written as s^3. For example, a cube with edges 3 cm long has volume 3 × 3 × 3 = 27 cm³.

A cuboid is like a box: its opposite faces are equal rectangles. If it has length l, breadth b and height h, then its volume is l × b × h. A cuboid 5 cm long, 4 cm wide and 2 cm high has volume 5 × 4 × 2 = 40 cm³.

A prism is a shape that has the same cross-section all the way through from one end to the other. The cross-section is the flat shape you see when you cut straight across the prism. A cuboid is one kind of prism, and so is a triangular prism, whose cross-section is a triangle. The volume of any prism is found the same way: the area of the cross-section multiplied by the length of the prism.

Volume of a prism = area of cross-section × length.

Worked example 1: Finding the volume of a cube
A cube has edges 4 cm long. Find its volume.

Step 1: A cube's volume is edge × edge × edge.
Step 2: 4 cm × 4 cm × 4 cm = 64 cm³.
Answer: The volume is 64 cm³.

Worked example 2: Finding the volume of a cuboid
A cuboid is 8 cm long, 5 cm broad and 3 cm high. Find its volume.

Step 1: A cuboid's volume is length × breadth × height.
Step 2: 8 cm × 5 cm × 3 cm = 120 cm³.
Answer: The volume is 120 cm³.

Worked example 3: Finding the volume of a triangular prism
A prism has a triangular cross-section with base 6 cm and height 4 cm, and the prism is 10 cm long. Find its volume.

Step 1: Find the area of the triangular cross-section. The area of a triangle is ½ × base × height, so ½ × 6 cm × 4 cm = 12 cm².
Step 2: Multiply the cross-section area by the length of the prism: 12 cm² × 10 cm = 120 cm³.
Answer: The volume is 120 cm³.

Practice

1. A cube has edges 2 cm long. Find its volume.
2. A cuboid is 6 cm long, 4 cm broad and 5 cm high. Find its volume.
3. A triangular prism has a triangular cross-section whose base is 8 cm and height is 5 cm. The prism is 9 cm long. Find its volume.
4. Name one shape from this lesson that has only a single curved face and no flat faces, and say how many faces, edges and corners a cylinder has.

Check your work

Attempt every question before you read this page.

Cylinders and Spheres: Their Shapes and Properties

1. A cylinder has 2 faces (two flat circles), 2 curved edges, and 0 vertices.

   Count the flat circular top and bottom as the two faces. The curved side is not a flat face. Where the curved surface meets each circular face there is a curved edge, so there are 2 edges. There are no corners, so there are 0 vertices.
2. A beach ball is a sphere, because it is round all over with no flat faces and no edges. A cone and a coin have flat surfaces.

   A sphere has no flat faces and no edges, only one curved surface. A beach ball fits this. A coin has two flat circular faces, and a cone has a flat circular base, so neither is a sphere.
3. It is a cylinder.

   A cylinder has exactly two flat circular faces (top and bottom) joined by one curved surface. A sphere has no flat faces, so the solid described must be a cylinder.
4. A sphere has 0 flat faces, 0 edges, 0 vertices, and 1 curved surface.

   A sphere is perfectly round all over. It has no flat face anywhere, so 0 faces; no line where surfaces meet, so 0 edges; no corners, so 0 vertices; and its entire outside is one curved surface.
5. For example, a drinking glass is a cylinder and an orange is a sphere. (Any similar correct examples are accepted.)

   A cylinder has a flat circular top and bottom joined by a curved surface (e.g. a glass or a tin). A sphere is round all over with no flat face (e.g. an orange, a football).

Cylinders, Spheres, and Finding Volume

1. 8 cm³

   Volume of a cube = edge × edge × edge = 2 cm × 2 cm × 2 cm = 8 cm³.
2. 120 cm³

   Volume of a cuboid = length × breadth × height = 6 cm × 4 cm × 5 cm = 120 cm³.
3. 180 cm³

   Area of triangular cross-section = ½ × base × height = ½ × 8 cm × 5 cm = 20 cm². Then multiply by the prism length: 20 cm² × 9 cm = 180 cm³.
4. A sphere has only one curved face and no flat faces. A cylinder has two flat circular faces, one curved face, and no corners (and no edges that are corners, though the rims are circles).

A sphere is perfectly round, so its entire surface is a single curved face. A cylinder is like a tin or drum: its two ends are flat circles (two flat faces), the curved side is one more face, and because all the faces meet along smooth curves rather than sharp points, it has no corners.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-ff3df83f	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____